

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

850 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

850 KWP GRID-TIED SOLAR PLANT

Net investment ₹3.08 Cr after input tax credit, generating 13,03,050 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹6.21 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	850 kWp
Year 1 generation	13,03,050 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹93.30 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹3.23 Cr
GST at 5% — recovered as ITC	₹15.38 L
Net capitalised cost	₹3.08 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹77.42 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	3,07,61,905	1,23,04,762	1,84,57,143	30,96,863
2	1,84,57,143	73,82,857	1,10,74,286	18,58,117
3	1,10,74,286	44,29,714	66,44,571	11,14,870
4	66,44,572	26,57,829	39,86,743	6,68,922
5	39,86,743	15,94,697	23,92,046	4,01,353
6	23,92,046	9,56,818	14,35,227	2,40,812
7	14,35,227	5,74,091	8,61,136	1,44,487
8	8,61,137	3,44,455	5,16,682	86,692
9	5,16,682	2,06,673	3,10,009	52,015
10	3,10,009	1,24,004	1,86,005	31,209
11	1,86,005	74,402	1,11,603	18,725
12	1,11,603	44,641	66,962	11,235
Tax shield, years 1-12				77,25,303

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30-year post-tax cash flow

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How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	13,03,050	93,29,838	—	1,23,04,762	(7,48,729)	1,00,78,567	1,00,78,567
2	12,96,535	96,54,516	5,10,000	73,82,857	4,43,374	87,01,142	1,87,79,709
3	12,90,052	99,90,494	5,10,000	44,29,714	12,71,180	82,09,313	2,69,89,022
4	12,83,602	1,03,38,163	5,10,000	26,57,829	18,04,630	80,23,533	3,50,12,555
5	12,77,184	1,06,97,931	5,10,000	15,94,697	21,62,745	80,25,186	4,30,37,741
6	12,70,798	1,10,70,219	5,10,000	9,56,818	24,16,984	81,43,235	5,11,80,976
7	12,64,444	1,14,55,462	5,10,000	5,74,091	26,10,267	83,35,196	5,95,16,172
8	12,58,122	1,18,54,112	5,10,000	3,44,455	27,68,394	85,75,719	6,80,91,891
9	12,51,831	1,22,66,636	5,10,000	2,06,673	29,06,895	88,49,741	7,69,41,632
10	12,45,572	1,26,93,514	5,10,000	1,24,004	30,35,138	91,48,377	8,60,90,009
11	12,39,344	1,31,35,249	5,10,000	74,402	31,58,797	94,66,452	9,55,56,461
12	12,33,147	1,35,92,355	5,10,000	44,641	32,81,332	98,01,024	10,53,57,485
13	12,26,982	1,40,65,369	5,10,000	26,785	34,04,874	1,01,50,495	11,55,07,980
14	12,20,847	1,45,54,844	5,10,000	16,071	35,30,762	1,05,14,083	12,60,22,063
15	12,14,742	1,50,61,353	5,10,000	9,643	36,59,858	1,08,91,495	13,69,13,558
16	12,08,669	1,55,85,488	5,10,000	5,786	37,92,743	1,12,82,745	14,81,96,303
17	12,02,625	1,61,27,863	5,10,000	3,471	39,29,830	1,16,88,033	15,98,84,336
18	11,96,612	1,66,89,113	5,10,000	2,083	40,71,435	1,21,07,678	17,19,92,014
19	11,90,629	1,72,69,894	5,10,000	1,250	42,17,816	1,25,42,078	18,45,34,092
20	11,84,676	1,78,70,886	5,10,000	750	43,69,199	1,29,91,687	19,75,25,779
21	11,78,753	1,84,92,793	5,10,000	450	45,25,796	1,34,56,997	21,09,82,776
22	11,72,859	1,91,36,342	5,10,000	270	46,87,810	1,39,38,532	22,49,21,308
23	11,66,995	1,98,02,287	5,10,000	162	48,55,442	1,44,36,845	23,93,58,153
24	11,61,160	2,04,91,406	5,10,000	97	50,28,896	1,49,52,510	25,43,10,663
25	11,55,354	2,12,04,507	5,10,000	58	52,08,379	1,54,86,128	26,97,96,791
26	11,49,577	2,19,42,424	5,10,000	35	53,94,104	1,60,38,320	28,58,35,111
27	11,43,829	2,27,06,020	5,10,000	21	55,86,289	1,66,09,731	30,24,44,842
28	11,38,110	2,34,96,190	5,10,000	13	57,85,161	1,72,01,029	31,96,45,871
29	11,32,420	2,43,13,857	5,10,000	8	59,90,953	1,78,12,904	33,74,58,775
30	11,26,757	2,51,59,980	5,10,000	5	62,03,906	1,84,46,074	35,59,04,849
Total	3,63,85,277	48,00,49,105	1,47,90,000	3,07,61,901	10,93,54,260	35,59,04,849	35,59,04,849

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

850 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹6.21 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹35.59 Cr
30-year generation	3,63,85,276 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 3,63,85,276 units, avoiding 29,835.9 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹3.23 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 850 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.